MODULE 42 · EXAM STRATEGY - ALL SECTIONS

How PDE Questions Are Written

The anatomy of a question, the qualifier words and what they eliminate, Google-recommended practice, numbered step options, choose-two technique, and pacing.

SECTIONS All five MODULE 42 of 45 SCENARIOS 4

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Every question is a design brief with one constraint hidden in it. **Find the constraint, and most options fall away.**

Four parts of every question

PART	WHAT IT HOLDS	WHAT TO DO
Setting	The company, its systems, the volumes	Note the facts that are fixed
Constraint	The qualifier: cheapest, fastest, least effort	Underline it, mentally
Ask	The last sentence: what should they do?	Read it first
Options	One right, one nearly right, two wrong	Strike, then compare two

THE PATTERN

The nearly right option works. It just breaks the constraint. That is the one to beat.

What each qualifier eliminates

THE QUESTION SAYS	STRIKE OUT
Least operational overhead	VMs, clusters you size, custom code you maintain
Most cost-effective	Premium tiers, always-on capacity, extra copies
Minimise latency, near real time	Nightly batches, hourly schedules, manual steps
Without changing code	Rewrites into another framework or language
Sensitive data, comply	Raw copies, broad roles, data leaving the boundary
Minimal data loss	Nightly backups, asynchronous-only designs

THE PAYOFF

One qualifier can strike two options before you think about the products at all.

How the questions have shifted

QUALIFIER IN THE STEM	OLDER HALF	NEWER HALF
Security, compliance, personal data	17	31
Reliability, RPO, outages	7	15
Managed, least operational effort	6	14
Google-recommended practice	3	9
Choose two or three	18	3

HONEST LIMIT

Counts from the practice bank used to plan this course, not from the live exam. The direction is what matters.

What Google-recommended means

- **Managed** over self-managed. **Serverless** over clusters you size.
- A **built-in feature** over custom code that does the same job.
- **Predefined roles** on **groups**, at the narrowest level that works.
- Never a **retired** product, a shared key, or a basic role like Editor.

> THE TIEBREAKER
>
> When two options both work, the recommended one is usually the one with less to look after.

Five steps through any question

01

Read the ask

The last sentence first

02

Find the qualifier

Cheapest, fastest, least effort

03

Strike the breakers

Any option that breaks a fixed fact

04

Compare the two left

On the qualifier alone

05

Pick the simplest

That still meets everything

REMEMBER

If you are comparing four options at the end, you skipped a step.

Nearly right or right?

NEARLY RIGHT

Works, but breaks one thing

- Meets the goal, costs too much
- Or needs a team to run it
- Or loses data the question protects
- Often the most impressive option

RIGHT

Meets every constraint

- Meets the goal and the qualifier
- Respects every fixed fact
- Usually managed, often simpler
- Sometimes the least exciting option

THE TRAP

The most powerful product is wrong when nobody asked for its power.

Options written as numbered steps

- Each option is a **sequence**. One bad step sinks the whole option.
- Options often share most steps. **Compare only the steps that differ.**
- **Order matters**: snapshot before an update, test before production.
- Watch for a step that is **unnecessary**, or a product that is retired.

> THE PATTERN
>
> Read the steps that differ first. Often the question is decided by a single line.

Choose two

- The question tells you **how many** to select. Pick exactly that many.
- Judge each option **on its own**: true or false, against the setting.
- The right pair usually **complements**: one part each, such as storage and access.
- Two options that are **alternatives** to each other? At most one is right.

> REMEMBER
>
> Choose-two is rarer now, but it still appears. Treat it as several true-or-false questions.

Pacing and the clock

- 2 hours for **40 to 50** questions: about **2.5 minutes** each.
- First pass: answer what you can, and **flag** the slow ones.
- Second pass: return to flagged questions with the time left.
- A blank scores nothing. **Answer every question**, even a guess.

> THE NUMBERS
>
> A question taking five minutes is costing you two others. Flag it and move on.

What the exam is not

- **No case studies** in the current guide. Old Flowlogistic and MJTelco material is outdated.
- **No labs**. Every question is multiple choice or multiple select.
- **No published pass mark**. Figures quoted online are guesses.
- **Old product names** are likely, as module 40 showed.

> CURRENCY CHECK
>
> Prepare for the exam Google describes today, not the one older courses describe.

Setting phrases that carry a hint

THE SETTING SAYS	IT USUALLY POINTS TO
A **small team**, no operations staff	Serverless and managed services
Existing Spark, Kafka or SQL code	Keep it: Managed Spark, Managed Kafka, BigQuery
Data **must stay** in a country	Location choice, perimeters, residency checks
Analysts who know SQL	BigQuery, Dataform, BigQuery ML
Within minutes of the event	Streaming, not a schedule
Once a day is enough	Batch loads, which are free

THE PATTERN

The setting is never decoration. Every fact in it is there to rule something out.

> Larkspur's three-person analytics team owns twelve existing PySpark jobs that run for about an hour each night.
>
> Nobody on the team wants to manage infrastructure. The jobs must run with the least operational overhead.

How should the jobs run?

A Spark on Compute Engine VMs, built with an instance template

B Spark on a GKE cluster, using the open-source Spark operator

C A Managed Spark cluster, kept running and sized for the peak

D Managed Spark serverless batches, with no cluster to size

◆ PAUSE AND DECIDE

How should the jobs run?

A Compute Engine

Every VM is the team's to patch and run.

B GKE operator

A Kubernetes cluster to run as well.

C Always-on cluster

Managed, but sized, and idle 23 hours a day.

D Serverless batches

Submit the job. Nothing to size or keep running.

THE SIGNAL

Existing Spark, least overhead, short nightly runs: serverless batches.

> The readings pipeline must get new code that renames one step. Larkspur wants a way back if the new code misbehaves.
>
> No readings may be lost, and the deduplication state must be kept.

Which sequence should the team follow?

A. 1. Cancel the job. 2. Launch the new code. 3. Seek Pub/Sub back to the cancel time.

B. 1. Take a Dataflow snapshot. 2. Update with a name mapping. 3. Roll back from it if needed.

C. 1. Update with a name mapping. 2. Take a Dataflow snapshot. 3. If needed, start from the snapshot.

D. 1. Drain the job. 2. Take a Dataflow snapshot. 3. Launch the new code from that snapshot.

◆ PAUSE AND DECIDE

Which sequence should the team follow?

A Cancel first

State is lost, and replays create duplicates.

B Snapshot, then update

Rollback point first, state kept by the update.

C Update, then snapshot

Too late: the rollback point comes after the change.

D Drain first

Draining discards the deduplication state.

THE SIGNAL

Numbered steps: order decides. The rollback point must come first.

> A new team of six analysts needs to query the curated dataset in BigQuery. They must not change data, and must not see other datasets.
>
> The security team asks for the Google-recommended approach.

How should access be granted?

A Data Viewer on the dataset to a group, plus Job User on the project

B The basic Viewer role on the whole project, to each analyst individually

C A shared service account key, with Data Viewer granted on the dataset

D A custom role copied from BigQuery Admin, granted on the project

◆ PAUSE AND DECIDE

How should access be granted?

A Group, dataset, predefined
Least privilege, at the narrowest level.

B Project Viewer
Sees every dataset, and granted per person.

C Shared key
A shared secret. Never recommended.

D Copied Admin role
Far too broad, and at project level.

THE SIGNAL

Google-recommended access: predefined roles, a group, the narrowest scope.

> Meter readings land in Cloud Storage as files every thirty minutes. Finance needs yesterday's totals in BigQuery by 07:00 each morning.
>
> The requirement is the lowest cost. Nothing needs to be fresher than that.

How should the data reach BigQuery?

A. A streaming Dataflow job, writing through the Storage Write API

B. BigQuery continuous queries, running on a dedicated reservation

C. A nightly scheduled load job from Cloud Storage into BigQuery

D. Datastream, replicating the files into BigQuery as they arrive

◆ PAUSE AND DECIDE

How should the data reach BigQuery?

A Streaming Dataflow

Fresh, but workers run all day, every day.

B Continuous queries

Real-time capacity nobody asked for.

C Nightly load job

Free on the shared pool, done well before 07:00.

D Datastream

Replicates databases, not files.

THE SIGNAL

Once a day is enough, lowest cost: a batch load job.

Module 42 in five lines

- Read the **ask** first. Find the **qualifier**. It decides the answer.
- Strike options that break a **fixed fact**, then compare the two left.
- Google-recommended: **managed**, built-in, predefined roles on groups.
- Numbered steps: compare **only what differs**. Choose two: judge each alone.
- About **2.5 minutes** a question. Flag, return, and answer everything.

COMING UP

Next: module 43, practice paper one. Twenty new questions, mixed sections.